1. User Registration

The registration interface of the RGPAS platform is shown in Figure 1. Users are required to provide a valid username, password, and academic email address (e.g., ending with *.edu*, *.ac.cn*, *.ac.uk*). After filling in all required information, click the "Register the account" button to complete the registration process.

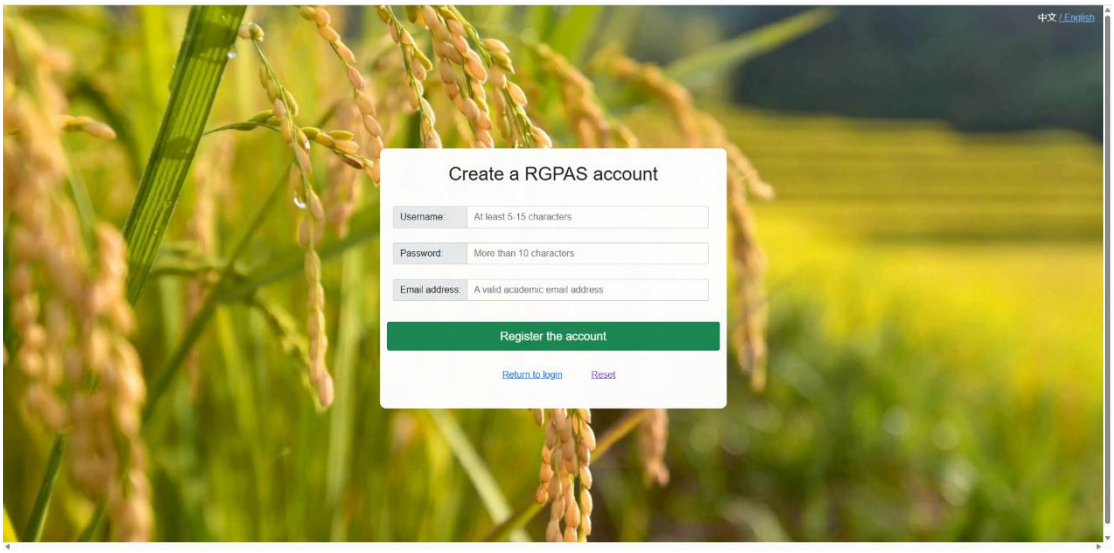


Figure 1 User registration interface

2. User Login

The login interface is illustrated in Figure 2. Registered users can log in by entering their verified email address and password, then clicking the "User Login" button to access the analysis dashboard. If the email or password is incorrect, an error message will be displayed.

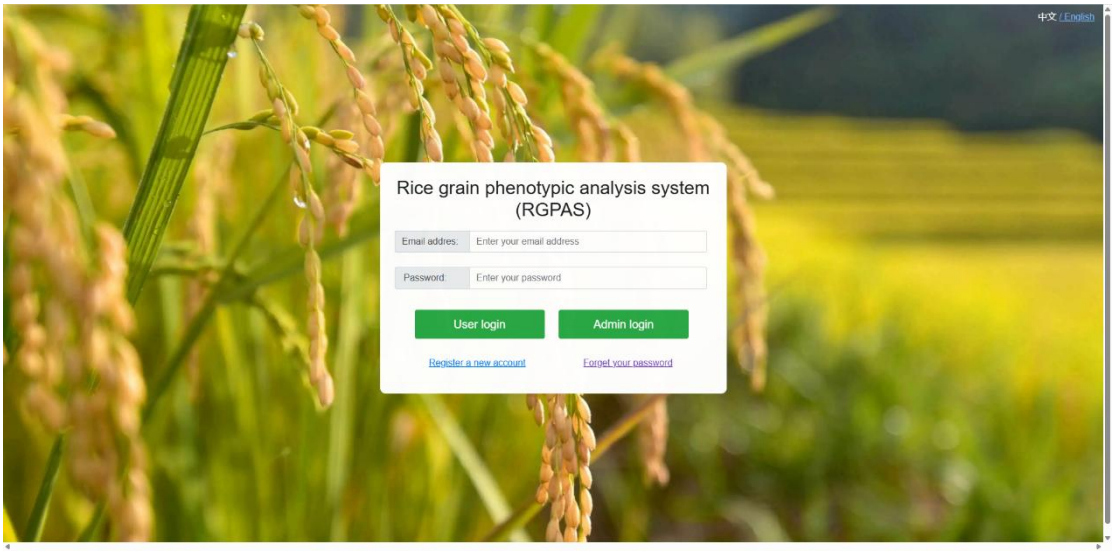


Figure 2 User login interface

3. Analyse a single image

After successful login, users are directed to the "Analyse a Single Image" module by default (Figure 3). Users can download the test images through the "Download test images" module.

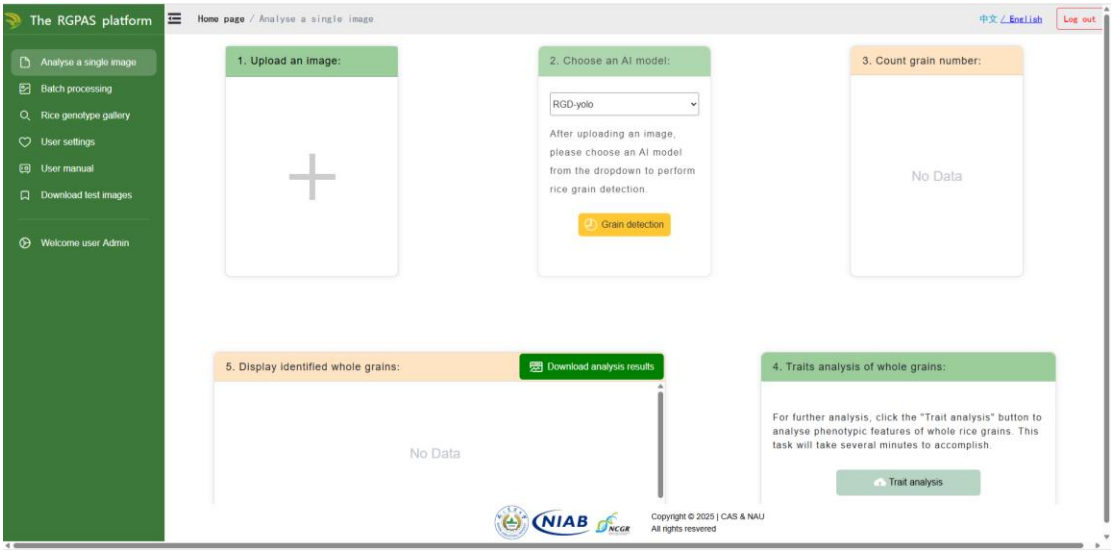


Figure 3 Interface of the "Analyse a Single Image" mode

This module consists of five functional sections:

1. Upload an image

   Click the "+" button to upload a rice panicle image for analysis.

2. Choose an AI model

   Select an AI detection model from the dropdown menu. RGPAS provides three models for panicle grain detection: RGD-YOLO, YOLOv8, and YOLOv10. After model selection, click "Grain detection" to initiate detection.

3. Display grain detection results

   Once processed, the detected grains are visualised in the "Count grain number" panel. Users can click the image to view a magnified version of the detection output.

4. Traits analysis of whole grains

   By clicking the "Traits analysis" button, the backend automatically runs the EfficientNet-RG model to classify and extract whole grain objects for further trait computation.

5. Display identified whole grains

   The extracted full grains from the uploaded image will be displayed in this panel upon completion.

6. Download analysis results

   Click "Download analysis results" to export the processed data, including grain counts, trait metrics, and visual outputs.

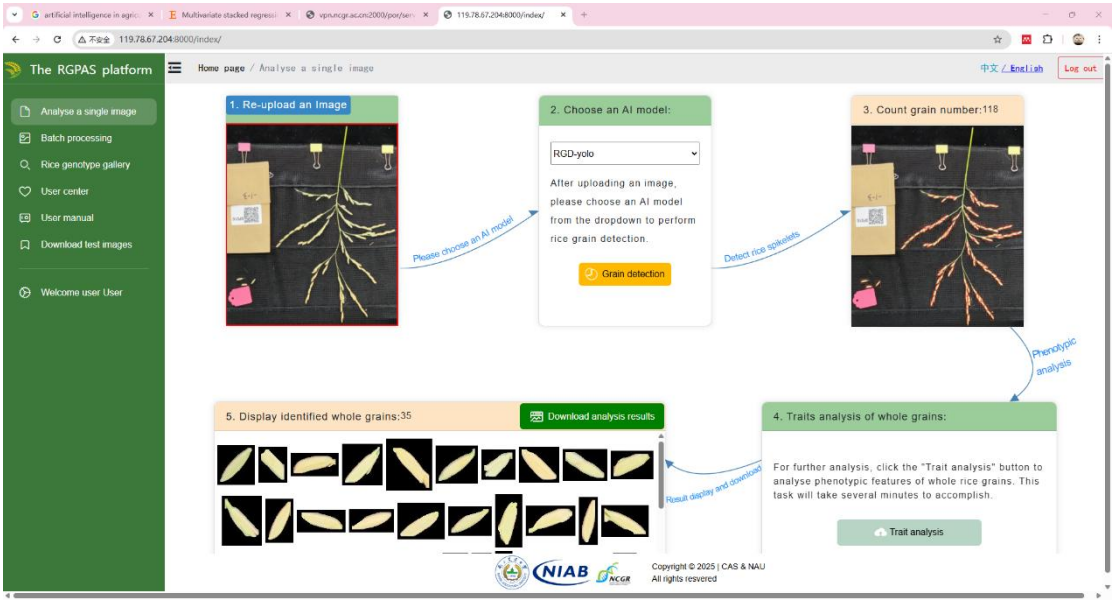


Figure 4 Single image completion view

4. Batch processing

Click “Batch processing” from the left navigation bar to access the batch analysis interface (Figure 5)

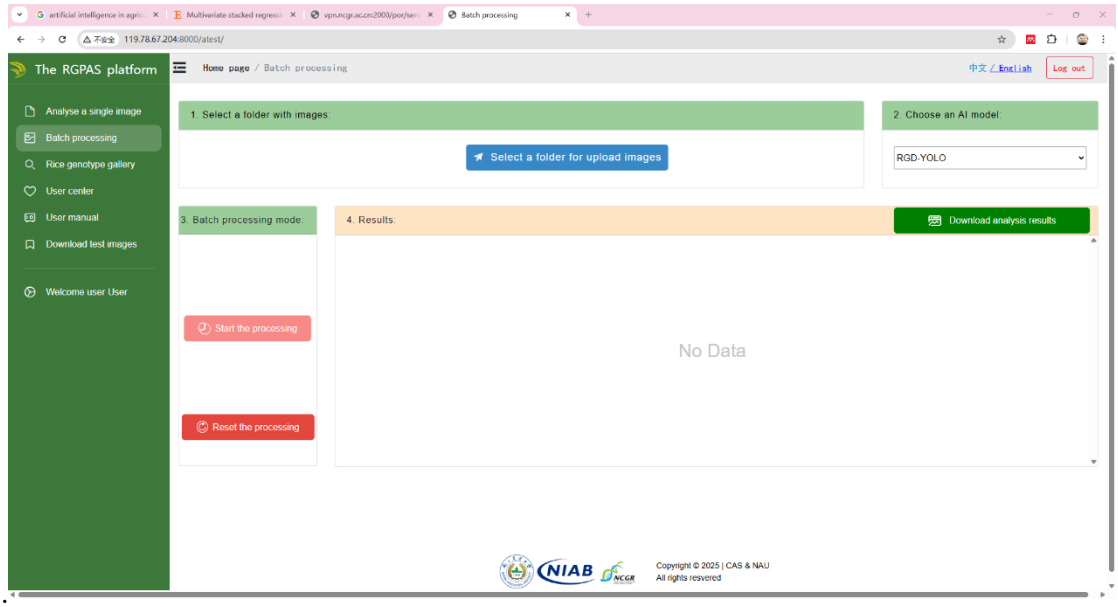


Figure 5 Batch processing

This module enables high-throughput analysis of multiple images simultaneously. It includes four key components:

1. Select a folder with images

   Choose a local folder containing the rice panicle images to be analysed.

2. Choose an AI model

   Select the appropriate detection model (e.g., RGD-YOLO, YOLOv8, or YOLOv10).

3. Batch processing mode

   Click “Start the processing” to begin automatic batch analysis.

4. Results

   Once analysis is completed, results are displayed in the “Results” window. A visual table summarises the processed images, including ID; image name; uploaded image; processed image; No. of grains; No. of identified whole grains and images of whole grains.

5. Download analysis results

   Click “Download analysis results” to download the full batch-level results as ZIP files.

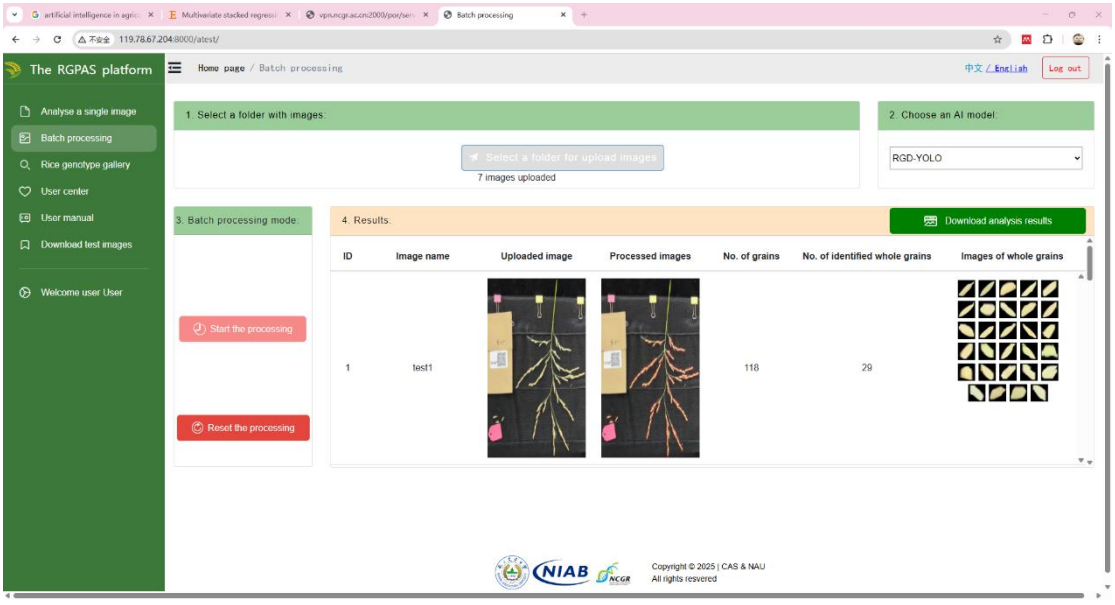


Figure 6 Batch analysis completion view

5. Rice genotype gallery

Click “Rice genotype gallery” on the navigation bar to access the integrated a digital rice germplasm database (Figure 7). This gallery hosts high-resolution images and phenotypic data for 916 landraces and 176 recombinant inbred lines (RILs). Users can search for a specific accession by entering its identifier in the Information box.

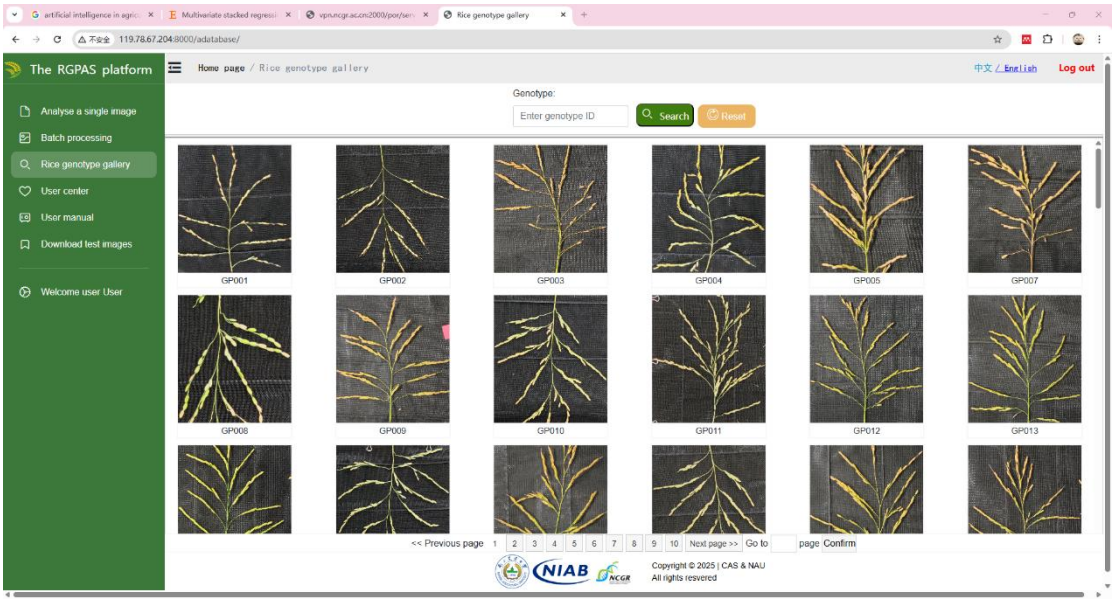


Figure 7 Rice genotype gallery

Clicking on any target image redirects to a detail page (Figure 8) containing:

- Grain detection and trait analysis results
- A detailed description of the selected accession
- Download options for the full phenotypic whole grain dataset of that accession via the “Download phenotypic datasets” button

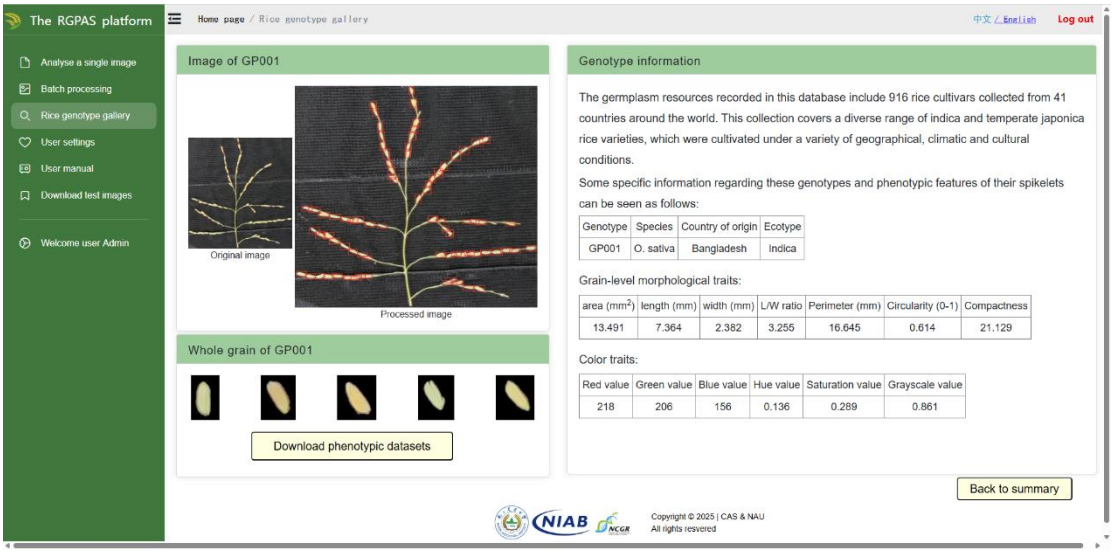


Figure 8 Detail view of a rice genotype

6. User setting

Users can manage their profile information through the “User Setting” section (Figure 9). Within this page, you can edit personal details, reset your password, or update contact information associated with your account.

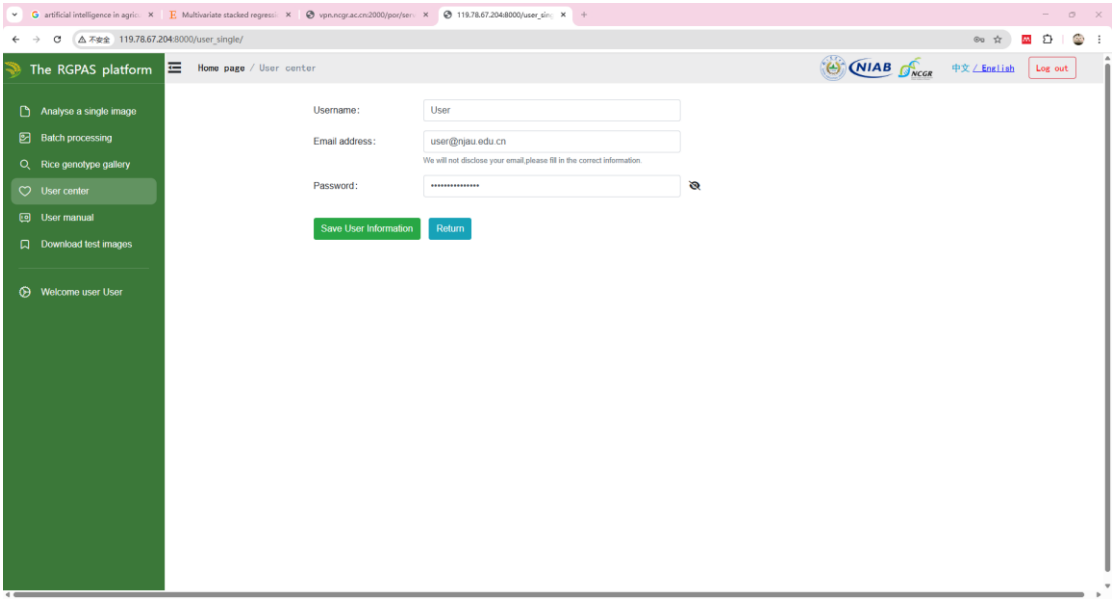


Figure 9 User settings interface